INCH-POUND
MIL-DTL-24243D
19 April 2023
SUPERSEDING
MIL-R-24243C
w/AMENDMENT 1
30 September 1994

DETAIL SPECIFICATION

RIVETS, BLIND, NONSTRUCTURAL, RETAINED MANDREL, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense (DoD).

1 SCOPE

1.1 Scope. This specification covers the requirements for pull-stem expandable blind rivets. Each rivet consists of a body assembled on a mandrel for fastening where access is available to one side only.

1.2 Classification. Rivets should be furnished in the head styles, material combinations, and open or closed-end, as specified in the applicable specification sheets (see 3.1).

2 APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to U.S. Army DEVCOM Ground Vehicle Systems Center, ATTN: Standardization, FCDD-GVS-SAT, MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to usarmy.detroit.devcom-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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FEDERAL SPECIFICATIONS

QQ-A-430	-	Aluminum Alloy Rod and Wire; for Rivets and Cold Heading
QQ-N-281	-	Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip Wire, Forgings, and Structural and Special Shaped Sections

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5541	-	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-DTL-16232	-	Phosphate Coating, Heavy, Manganese or Zinc Base
MIL-PRF-8625	-	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-PRF-32647	-	Zinc-Nickel Electroplating for Fasteners

(See ASSIST database for list of specification sheets.)

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A313/A313M	-	Standard Specification for Stainless Steel Spring Wire
ASTM A493	-	Standard Specification for Stainless Steel Wire and Wire Rods for Cold Heading and Cold Forging
ASTM A510/A510M	-	Standard Specification for General Requirements for Wire Rods and Coarse Round Wire, Carbon Steel, and Alloy Steel
ASTM A580/A580M	-	Standard Specification for Stainless Steel Wire
ASTM B211/B211M	-	Standard Specification for Aluminum and Aluminum-Alloy Rolled or Cold Finished Bar, Rod, and Wire
ASTM B633	-	Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel
ASTM D3951	-	Standard Practice for Commercial Packaging

(Copies of these documents are available online at https://www.astm.org.)

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NASM1312-8	-	Fastener Test Methods, Method 8, Tensile Strength
NASM1312-20	-	Fastener Test Methods, Method 20, Single Shear
NASM1515	-	Fastener Systems for Aerospace Applications

(Copies of these documents are available online at https://www.aia-aerospace.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://www.asq.org.)

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SAE INTERNATIONAL

SAE AMS2700	-	Passivation of Corrosion Resistant Steels
SAE AMS-QQ-W-428	-	Wire, Steel, Carbon, (High Carbon, Round, for Mechanical Springs, General Purpose)

(Copies of these documents are available online at https://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern.

3.2 Materials. The materials used shall be such as to produce bodies and mandrels in compliance with the requirements of this specification and as specified in the applicable specification sheet and Table I.

3.2.1 Aluminum alloys.

a. Type 1100 (UNS A91100), 5052 (UNS A95052), and 5056 (UNS A95056) aluminum alloys shall conform to the applicable requirements of QQ-A-430.
b. Type 2024 (UNS A92024) aluminum alloy shall conform to the applicable requirements of ASTM B211/B211M.

3.2.2 Carbon steel. Carbon steel shall be of the grade required by the applicable specification sheet.

3.2.3 Carbon steel wire. Carbon steel wire employed in the fabrication of mandrels shall conform to the requirements of ASTM A510/A510M or SAE AMS-QQ-W-428 as necessary to install rivet bodies to meet the applicable strength requirements of Table I.

3.2.4 Nickel-copper alloy. Nickel-copper alloy bodies shall be in accordance with Class A of QQ-N-281.

3.2.5 Corrosion resistant steel. Corrosion resistant steel employed in the manufacture of bodies and mandrels shall be of the type specified in the applicable specification sheet and in accordance with ASTM A493, ASTM A580/A580M or ASTM A313/A313M at the supplier's option.

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Table I. Material combination symbols.

Symbol	Material Combination	Rivet Dia	Minimum Shear Load Lbs	Minimum Tension Load Lbs
MIL-R-24243/1	(Open-End, Domed Head)			
A	Body: 5056 Al Alloy (UNS A95056) Mandrel: 1006-1038 Steel (UNS G10060-G10380)	.094 .125 .156 .188 .250	90 170 260 380 700	120 220 350 500 920
B	Body: 5052 Al Alloy (UNS A95052) Mandrel: 2024 Al Alloy (UNS A92024)	.094 .125 .156 .188	70 120 190 260	80 150 230 320
D	Body: 1006-1010 Steel (UNS G10060-G10100) Mandrel: 1030-1060 Steel (UNS G10300-G10600)	.094 .125 .156 .188 .250	130 260 370 540 1000	170 310 470 680 1240
F	Body: 302 CRES (UNS S30200) 302-HQ-CRES (UNS S30430) 305 CRES (UNS S30500) Mandrel: 305-431 CRES (UNS S30500-S43100)	.125 .188	420 950	530 1200
G	Body: 302 CRES (UNS S30200) 302-HQ-CRES (UNS S30430) 305 CRES (UNS S30500) Mandrel: 1030-1060 Steel (UNS G10300-G10600)	.125 .188	420 950	530 1200
MIL-R-24243/2	(Open-End, Domed Head)			
E	Body: Nickel Copper Alloy (UNS N04400) Mandrel: 1030-1060 Steel (UNS G10300-G10600)	.125 .156 .188 .250	350 550 800 1400	450 700 1000 1850
MIL-R-24243/3	(Open-End, Large Domed Head)			
A	Body: 5056 Al Alloy (UNS A95056) Mandrel: 1006-1038 Steel (UNS G10060-G10380)	.125 .188	170 380	220 500
B	Body: 5052 Al Alloy (UNS A95052) Mandrel: 2024 Al Alloy (UNS A92024)	.125 .188	120 260	150 320
D	Body: 1006-1010 Steel (UNS G10060-G10100) Mandrel: 1030-1060 Steel (UNS G10300-G10600)	.125 .188	260 540	310 680

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Table I. Material Combination Symbols – Continued.

Symbol	Material Combination	Rivet Dia	Minimum Shear Load Lbs	Minimum Tension Load Lbs
MIL-R-24243/4	(Open End, 100° Countersunk Head)			
A	Body: 5056 Al Alloy (UNS A95056) Mandrel: 1006-1038 Steel (UNS G10060-G10380)	.125	170	220
B	Body: 5052 Al Alloy (UNS A95052) Mandrel: 2024 Al Alloy (UNS A92024)	.125	120	150
MIL-R-24243/5	(Open End, 120° Countersunk Head)			
A	Body: 5056 Al Alloy (UNS A95056) Mandrel: 1006-1038 Steel (UNS G10060-G10380)	.125 .156 .188	170 260 380	220 350 500
B	Body: 5052 Al Alloy (UNS A95052) Mandrel: 2024 Al Alloy (UNS A92024)	.125	120	150
D	Body: 1006-1010 Steel (UNS G10060-G10100) Mandrel: 1030-1060 Steel (UNS G10300-G10600)	.125	260	310
E	Body: Nickel Copper Alloy (UNS N04400) Mandrel: 1030-1060 Steel (UNS G10300-G10600)	.125 .156 .188	350 550 800	450 700 1000
MIL-R-24243/6	(Closed End, Domed Head, Hollow Core)			
A	Body: 5056 Al Alloy (UNS A95056) Mandrel: 1012-1025 (UNS G10120-G10250) 1045-1075 Steel (UNS G10450-G10750)	.125 .156 .188 .250	240 350 500 900	280 480 690 1100
C	Body: 1100 Al Alloy (UNS A91100) Mandrel: 2024 Al Alloy (UNS A92024)	.125 .156 .188	100 130 210	110 160 250
MIL-R-24243/6	(Closed End, Domed Head, Filled Core)			
A	Body: 5056 Al Alloy (UNS A95056) Mandrel: 1012-1025 (UNS G10120-G10250) 1045-1075 Steel (UNS G10450-G10750)	.125 .156 .188	320 590 790	280 480 690

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Table I. Material Combination Symbols – Continued.

Symbol	Material Combination	Rivet Dia	Minimum Shear Load Lbs	Minimum Tension Load Lbs
MIL-R-24243/7	(Closed End, 120° Countersunk Head, Hollow Core)			
A	Body: 5056 Al Alloy (UNS A95056) Mandrel: 1012-1025 (UNS G10120-G10250) 1045-1075 Steel (UNS G10450-G10750)	.125 .156 .188	240 350 500	280 480 690
MIL-R-24243/7	(Closed End, 120° Countersunk Head, Filled Core)			
A	Body: 5056 Al Alloy (UNS A95056) Mandrel: 1012-1025 (UNS G10120-G10250) 1045-1075 Steel (UNS G10450-G10750)	.125 .156 .188	320 590 790	335 525 680
MIL-R-24243/8	(Open End, Snap Head)			
A	Body: 5052 Al Alloy (UNS A95052) Mandrel: 1016-1025 Steel (UNS G10160-G10250)	.125 .156 .188	165 255 345	230 375 525
MIL-R-24243/9	(Open End, 120° Countersunk Head)			
A	Body: 5052 Al Alloy (UNS A95052) Mandrel: 1016-1025 Steel (UNS G10160-G10250)	.125 .156 .188	155 225 350	205 300 530
MIL-R-24243/10	(Open End, Domed Head)			
A	Body: 5052 Al Alloy (UNS A95052) Mandrel: 1016-1025 Steel (UNS G10160-G10250)	.125 .156 .188	165 255 345	230 375 525

3.3 Surface finish.

3.3.1 Aluminum alloy rivets. Aluminum alloy rivets shall be furnished chemically treated in accordance with MIL-DTL-5541, Type II, Class 1A, or anodically treated in accordance with MIL-PRF-8625, Type IC, II or IIB. Chromate containing sealers shall be prohibited. No finish is required for closed-end rivets.

3.3.2 Carbon steel bodies and mandrels. Carbon steel bodies and mandrels shall be zinc plated in accordance with ASTM B633, Fe/Zn 5, Type V or VI, or phosphate coated in accordance with Type Z, Class 2 or 4 of MIL-DTL-16232, as specified (see 3.1). No finish is required for closed-end rivets.

3.3.3 Nickel-copper alloy bodies. Nickel-copper alloy bodies shall be zinc plated in accordance with ASTM B633, Fe/Zn 5, Type V or VI.

3.3.4 Corrosion resistant steel. Corrosion resistant steel (CRES) bodies and mandrels shall be passivated in accordance with SAE AMS2700, Method 1, Types 6-8, or Method 2. Passivation treatments containing hexavalent chromium shall be prohibited.

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3.4 Construction. Fastener assembly shall be as specified in the applicable specification sheet.

3.4.1 Rivets. Rivets shall consist of two parts, a body and a mandrel assembled in combination as specified on the applicable specification sheet (see 6.2).

3.4.2 Rivet bodies. Rivet bodies shall be capable of being expanded mechanically by the action of the mandrel when pulled into the rivet body as shown on Figure 1.

3.4.3 Dimensions. Dimensions shall be as specified in the applicable specification sheet.

3.4.4 Strength. Single shear and tension load capabilities of installed rivets shall not be less than specified in Table I.

3.5 Workmanship. Rivets shall be of uniform quality and free from injurious seams and other injurious defects.

4 VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use their own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptable of defective material.

4.2 Quality conformance inspection. Quality conformance inspections shall be as specified in Table II.

TABLE II. Quality conformance inspection.

INSPECTION	REQUIREMENT PARAGRAPH	EXAMINATION OR PARAGRAPH TEST
Dimensions	3.4.3	4.2.3
Protective Surface	3.3	4.3.3
Surface Treatment	3.3.4	4.3.4
Single Shear	3.4.4	4.3.1.1
Tension	3.4.4	4.3.1.2

4.2.1 Inspection lot. An inspection lot shall consist of rivet body and mandrel assemblies of the same material, finish and nominal size produced by the same manufacturer under essentially the same conditions and submitted for acceptable at one time.

4.2.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for inspection. Resubmitted lots shall be

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inspected using tightened inspection. Such lots shall be separated from new lots, and shall be clearly identified as reinspected lots.

4.2.3 Sampling for visual and dimensional examination. A random sample of body and mandrel assemblies shall be taken from each lot in accordance with ASQ Z1.4 as follows:

Major - Inspection level S3
Minor - Inspection level S2

Acceptance = 200 Rejection = 1

4.2.4 Sample for shear strength and tension loads. Sampling for shear strength and tension loads shall be in accordance with ASQ Z1.4 at Inspection Level S-2.

4.3 Methods of inspection.

4.3.1 Mechanical properties. Sample assemblies selected in accordance with 4.2.4 shall be tested as specified in 4.3.1.1 and 4.3.1.2 to assure compliance with Table I of this specification.

4.3.1.1 Single shear load. The shear test shall be performed in accordance with Test Method 20 of NASM1312-20, or equivalent.

4.3.1.2 Tension load. The tension test shall be performed in accordance with Test Method 8 of NASM1312-8, or equivalent.

4.3.2 Visual and dimensional examination. Samples of rivets shall be examined to verify conformance with this specification. Examination shall be conducted in accordance with Table III.

TABLE III. Classification of defects.

CATEGORY	DEFECT	INSPECTION METHOD
Critical	None defined	
Major		
101	Diameter of rivet body (see 3.5)	CIE /1
Minor		
201	Diameter of rivet body (see 3.4)	CIE
202	Length of rivet body	CIE
203	Thickness of rivet body head (see 3.4)	CIE
204	Diameter of mandrel (see 3.4)	CIE
205	Radius under head (where applicable) (see 3.4)	CIE
206	Length of mandrel (see 3.4)	CIE
207	Angle of rivet head (where applicable) (see 3.4)	CIE
208	Protective finish and surface treatment (see 3.3)	Visual
209	Workmanship (see 3.5)	Visual

1/ Commercial Inspection Equipment

4.3.3 Protective finish. Samples of aluminum alloy, carbon steel, and nickel-copper bodies and/or mandrels, shall be inspected for adequacy of protective finish in accordance with applicable specification of 3.3.

4.3.4 Surface treatment. Samples of corrosion resistant steel bodies and mandrels shall be passivated as specified in 3.3.4.

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4.3.5 Materials inspection. Materials inspections shall consist of certification supported by verifying data that the materials used in fabricating the body and mandrel assemblies are in accordance with the requirements of 3.2 prior to such fabrication.

5 PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with ASTM D3951 (see 6.2).

6 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Rivets required by this specification are intended for rapid and permanent fastenings substantially as shown on Figure 1. Rivet setting tools may be manually or power operated.

6.1.1 General. In general, these rivets are used to fasten metal to metal, wood to metal, and plastics to metal. Typical nonstructural applications are as follows: Attachments such as for handles, hinges, brackets and clips; lockers such as for food and clothes; bins, metal furniture, racks, shelves; assembly of tube sockets and terminal boards; fastening conduit clips, raceways, electrical boxes, distribution panels; fastening signs, bulletin boards, and safety guards around equipment; installing light fixture mounting brackets; repair of light sheet metal equipment.

6.1.2 Closed-end rivets. Closed-end rivets are intended for application (under the most optimum conditions, for example proper hole size, grip length and over coated with sealant) where gas, oil, water or air tightness is required.

6.1.3 Aerospace usage. These rivets are not intended for aerospace usage. For aerospace rivets, see NASM1515.

6.2 Ordering data. Acquisition requirements should specify the following:

a. Title, number, and date of this specification and applicable specification sheet.
b. Applicable specification sheet (see 3.1) part number.
c. Level (degree) of protection in accordance with ASTM D3951.

6.2.1 Acquisition. Acquisition of rivets manufactured prior to issuance of Revision D should not be used for new systems unless otherwise specified by the contract. This Revision (Revision D), is applicable to rivets manufactured after 14 March 2023.

6.3 Subject term (Key Word) listing.

Blind Fastener
Rivet, Blind
Rivet, Blind, Nonstructural

6.4 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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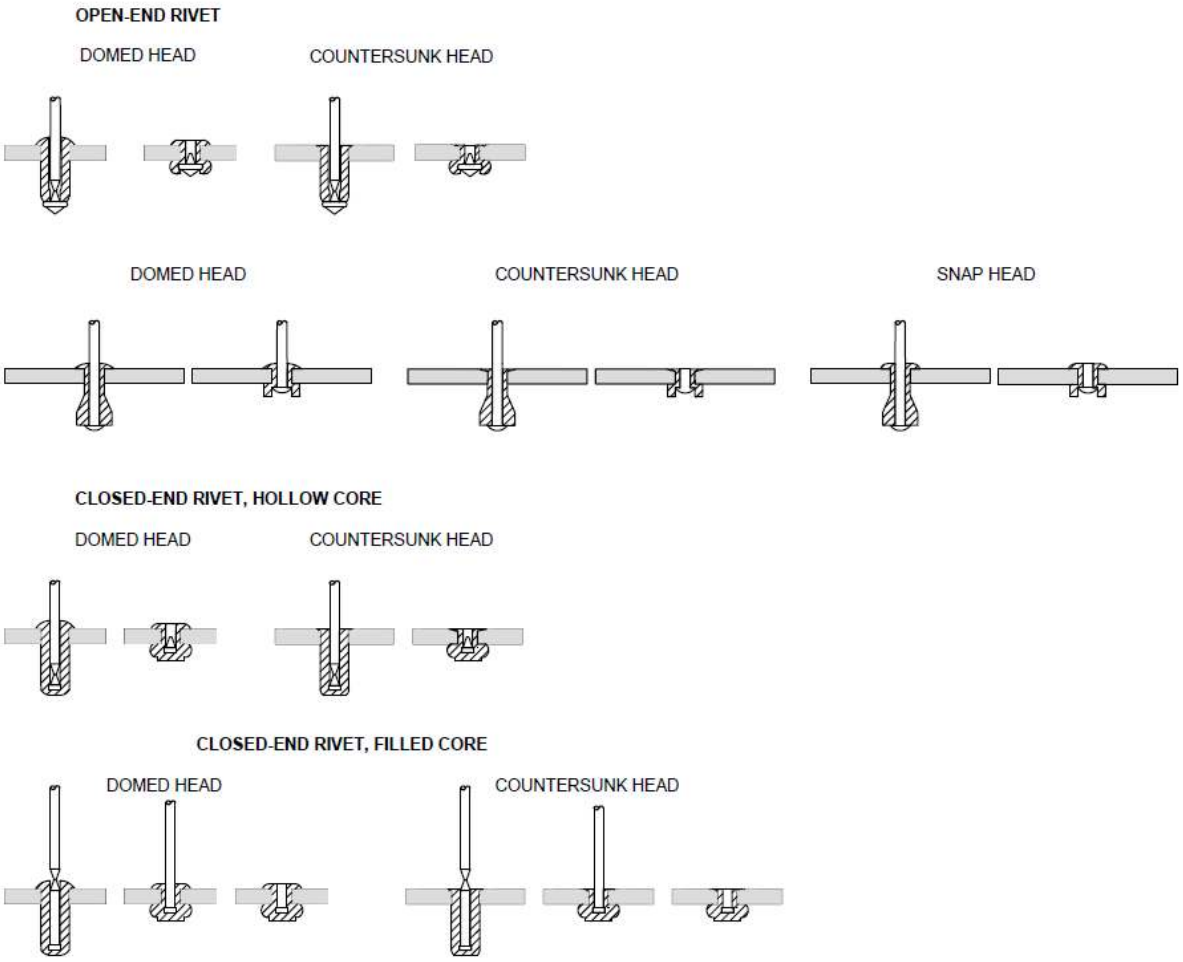


FIGURE 1. Illustration of rivet styles before and after setting.

Custodians:
Army - AR
Navy - AS
Air Force - 71
DLA - GS

Preparing Activity:
Army - AT

(Project 5320-2023-001)

Review Activities:
Army - AV, MI

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